

Name: AAVISHKAR SINGH AAVISHKAR SINGH | Reg. No: RA2511003011024

Unit “ 4 : Session “ 6 : SLO “ 2 | Hardwired Control “” Fill in the Blanks

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1. combinational logic (digital logic) “” control signals are generated by gate-level circuits.
2. decoder “” the instruction decoder decodes the opcode to initiate the appropriate control signals.
3. control memory “” hardwired control eliminates control-memory access time, making it faster than microprogrammed control.
4. timing (sequence/order of steps) “” the sequence counter determines which step of the instruction cycle is active.
5. combinational “” signals are produced by combinational logic design (with sequential elements for state).
6. control logic (instruction set) “” a major disadvantage is that the wired logic is difficult to modify.
7. sequence counter “” keeps track of the current step in instruction execution.
8. RISC (simple, fixed) “” hardwired control suits simple, fixed instruction set architectures.
9. clock (timing signal generator) “” timing signals are generated by the clock and step logic.
10. logic gates “” the entire control unit is built from logic gates and flip-flops.